

RYAN TANG

New York, New York

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Education

New York University

Expected Graduation: May 2027

B.A. Mathematics

New York, New York

Relevant Coursework: Machine Learning (Graduate), Honors Probability Theory, Mathematical Statistics, Real Analysis, Ordinary Differential Equations, Game Theory, Abstract Algebra, Modeling and Simulation, Linear Algebra, Calculus I - III, Data Structures

Experience

Undergraduate Researcher

May 2026 - Present

University of Minnesota - Twin Cities

Minneapolis, MN

- Engineered data pipelines to process large-scale geospatial and environmental simulation data on a shared HPC cluster
- Built a GRU-based spatio-temporal model for corn yield prediction from DayCent simulation outputs
- Extended model to a dual-output architecture, examining relationships between soil organic carbon and crop yields
- Presented baseline modeling results, informing next steps for model architecture and evaluation strategy

Data Analytics Intern

May 2025 - August 2025

Lewis and Tibbitts Inc.

San Jose, CA

- Worked directly with the COO to build Power BI dashboards analyzing business processes and operational performance
- Defined various KPIs to quantify foreman productivity, leading to a 10% increase in timeliness and efficiency
- Created monthly reports summarizing job site performance, cost efficiency, and schedule adherence for executive review

Undergraduate Researcher

June 2025 - August 2025

University of Nebraska Lincoln

Remote

- Studied nonlocal operator frameworks for modeling discontinuities, long-range interactions, and physical phenomena
- Implemented optimization-based algorithms to solve nonlocal inverse problems and recover unknown model parameters
- Developed nonlocal physics-informed neural networks incorporating integral operators to solve initial value problems

Information Technology Intern

West Valley Construction Company Inc.

May 2024 - August 2024

Campbell, CA

- Rebuilt the Employee Reimbursement Process workflow using Microsoft Power Automate to reduce manual input error
- Collaborated on interactive Power BI dashboards to visualize key performance metrics and operational efficiency

Projects

Heritability and Persistence of 2D Ising Systems | *Numerical Analysis*

- Explored the 2D Ising model, examining whether states are dominated by initial conditions or dynamical randomness.

GP + DL Hybrid Credit Scoring Model | *Machine Learning, Model Tuning*

- Implemented a hybrid machine learning pipeline using Genetic Programming and a Stacked Auto Encoder in credit risk

Climbing Movement Analysis | *Computer Vision, Python*

- Defined research backed quantifiable metrics to assess skill expression of climbing movements on the MoonBoard

Skills

Tools: Microsoft Azure, AWS, Google Cloud Platform (GCP), Unix/Linux, VMWare, Power Automate, Power BI, Tableau, Git/Github, Jupyter, VS Code, Anaconda

Programming Languages: Python, Java, C++, C, HTML/CSS, JavaScript/TypeScript, SQL, Powershell, R, Bash

Machine Learning & Data Science: NumPy, pandas, scikit-learn, TensorFlow, PyTorch, Matplotlib, Seaborn, SciPy, StatsModels, OpenCV, Data Wrangling, Model Evaluation, Statistical Inference

Leadership and Extracurricular

- NYU Climbing Competition Team Member
- NCAA DIII Water Polo for Occidental College (Fall 2023)
- Founder/Co-President of Oxy Volleyball and Oxy Bouldering
- High School Water Polo Captain and League MVP
- State Championship Qualified Swimmer and 14-time Section Qualifier